MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPE 5949A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, hydrogen

See figure 1

Mounting position: Any

Weight: 2.25 pounds (1.0 kg) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epy	epx	Ebb	Ip	egy	egx	Ecc	ib
Unit:	Vac	kv	kv	kVdc	Aac	v	v	Vdc	a
Maximum:	6.6	25.0 (Note 1)	25.0 (Note 2)	---	15	---	450	---	500
Minimum:	6.0	10.0	5% epy	5.0	---	Note 3	---	---	---
TEST CONDITIONS:	6.3	25.0	---	---	---	450	---	0	---

ABSOLUTE RATINGS:

Parameter:	Ib	prr	tj	Pb	tk	TA	dik/dt	Eres	Ires
Unit:	Adc	---	µs	---	---	°C	a/µs	Vac	Aac
Maximum:	0.5	---	0.005 (Note 4)	6.25×10^9	---	+75	2,500	5.5 (Note 5)	6.0
Minimum:	---	---	Note 3	---	180	-55	---	3.0	---
TEST CONDITIONS:	---	500	---	---	180	---	---	Note 6	---

GENERAL:

Qualification - Required

Ⓓ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
3241	Heater current	-		0.65	II	If	15.0	22.0	Aac
3241	Heater current	-	Eres = 4.5 Vac	0.65	II	Ires	2.0	5.0	Aac
(D) 3247	DC anode voltage for conduction	-	Eres = inscribed value -5 percent	0.65	II	Ebb	---	4,000	Vdc
3267	Instantaneous starting	8, 11	epy = 18.0 kv; Eres = inscribed value +5 percent	0.65	II	---	---	---	---
3246	Operation (1)	8	epy = 27.5 kv; prr = 450; Eres = inscribed value -5 percent; t = 30 minutes	0.65	II	egy	---	450	v
3246	Operation (2)	12, 13	epy 20 kv; prr = 1,200; Eres = 95 percent of optimum value	0.65	II	egy	---	450	v
(D) 3246	Operation of optimum reservoir voltage (1)	-	Operation (1)	0.65	II	Eres	3.0	5.5	Vac
(D) 3246	Operation of optimum reservoir voltage (2)	13	Operation (2)	0.65	II	Eres	3.0	5.5	Vac
3251	Pulse emission	-	ik = 500 a; tp = 5.0 µs ±10 percent; prr = 60 ±10 percent; tr = 0.5 µs (max); Eres = inscribed value; starting time of current pulse = 2.5 µs	0.65	II	egk	---	250	v
	Quality conformance inspection, part 2								
3246	Operation (3)	-	Operation (1), except Eres = inscribed value +5 percent; tk = 900; t = 1,200	---	---	egy	---	450	v
3246	Operation (4)	-	Operation (2), except Eres = 105 percent optimum value	---	---	egy	---	450	v
3256	Anode delay time	-	Operation (1); t = 120	---	---	tad	---	1.0	µs
3256	Anode delay time drift	14	Anode delay time	---	---	Δtad	---	0.25	µs
3261	Time jitter	15	Operation (1), except epy = 15.0 kv	---	---	tj	---	0.004	µs
	Quality conformance inspection, part 3								
---	Life test (1)	8	Group C; t = 96 hours "on" and 1 hour "off"; t = 500 hours	---	---	---	---	---	---
---	Life test (2)	13	Life test (1); operation (2); Zg = 250 ohms; t = 500 hours	---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Life test (3)	7, 8, 10	t = 500 hours	---	---	---	---	---	---
---	Life-test (1), (2), and (3) end points:								
3246	Operation (1)	-		---	---	egy	---	500	v
3247	DC anode voltage for conduction		egy = 500 v	---	---	Ebb	---	4,500	Vdc
3261	Time jitter	-	egy = 500 v; Zg = 250 ohms	---	---	tj	---	0.005	µs
3246	Operation at elevated ambient temperature	7, 8, 9	TA = 75°C; t = 5 hours	---	---	---	---	---	---

NOTES:

1. Instantaneous starting is not recommended. When it is absolutely necessary; however, the maximum permissible epy is 18.0 kv and shall not be attained in less than 0.04 second.
2. The peak inverse voltage, exclusive of a spike of 0.05 µs maximum duration, shall not exceed 5 kv during the first 25 µs following the anode pulse.
3. The driver pulse, measured at the tube socket with thyratron grid disconnected, shall be: egy = 550 volts minimum, 1,000 volts maximum; tr - 0.25 µs maximum; tp = 2.0 µs minimum; impedance of driver circuit 50 to 200 ohms.
4. Appreciably less jitter than 0.005 µs can be realized if anode voltage of 18 kv or more, maximum rated grid drive voltage and a grid impedance near the lower limit are used.
5. The optimum reservoir voltage for operation in accordance with operation (1) conditions is inscribed on the base of the tube and must be held to within ±5 percent.
6. Adjust reservoir voltage to value indicated on tube to within ±5 percent.
7. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.
8. The circuit constants shall be chosen so that at epy = 25.0 kv under resonant charging conditions: dik/dt = 2,500 a/µs minimum; ib = 500 a minimum; tp = 2.0 ±0.2 µs; prr - 500 minimum. The grid pulse characteristics shall be: tp = 2.0 µs maximum; tr = 0.25 µs minimum; internal impedance of driver - 250-ohms minimum.
9. This test shall be conducted for a total of 5 consecutive hours with no more than three kickouts and with no evidence of anode heating. The tube shall be started at Eres = 105 percent of the inscribed value and operate at this value for 4 hours. At the start of the fifth hour and while the tube is still operating, the reservoir voltage shall be lowered to 95 percent of the inscribed value and remain there for the final 1 hour of operation.
10. The life test shall be intermittent, with the tube operating approximately half the total elapsed time, and held without reservoir voltage or cathode-heater voltage approximately half the total time. The tube shall make a minimum of four starts during each 24-hour period. The life-test duration is measured in operating hours. Failure of the tube to operate following the minimum allowable cathode and reservoir heater warmup time constitutes failure of the test.
11. The tube shall operate satisfactorily on push-button starting within three attempts when the anode voltage (epy) is applied to the tube under test in such a manner as to rise from 0 to 18 kv within 0.03 second. (The filter in the rectifier shall be designed so that the epy reaches at least 9 kv within 0.015 second.) Any tube failing to start within three attempts will be considered a failure.
12. The anode circuit constants shall be so chosen that at resonant charging conditions; epy = 20 kv; ib = 200 a minimum; dik/dt = 2,500 a/µs minimum. Output pulse; tp = 1.0 µs ±10 percent; prr = 1,200 minimum; epx = 5.0 kv. Grid pulse the same as note 8. Adjust reservoir voltage to optimum value for these test conditions. The tube shall operate continuously for 20 minutes without continuous conduction.
13. The optimum reservoir voltage for operation in accordance with operation (2) conditions shall be so determined that the tube shall operate at 95 to 105 percent of the established reservoir voltage for these conditions.

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NOTES: -Continued

14. This test shall be performed simultaneously with the operation (1) test. An anode delay time measurement shall be made at the end of 2 minutes, 4 minutes, and 30 minutes of the operation (1) test. The Δtad measurement is the numerical difference between the 2-minute and 4-minute, or 2-minute and 30-minute anode delay time readings, whichever is greater.

15. The tj shall be not greater than the amount specified after 120 seconds of anode operation.

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Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 17, 80
DSA - ES

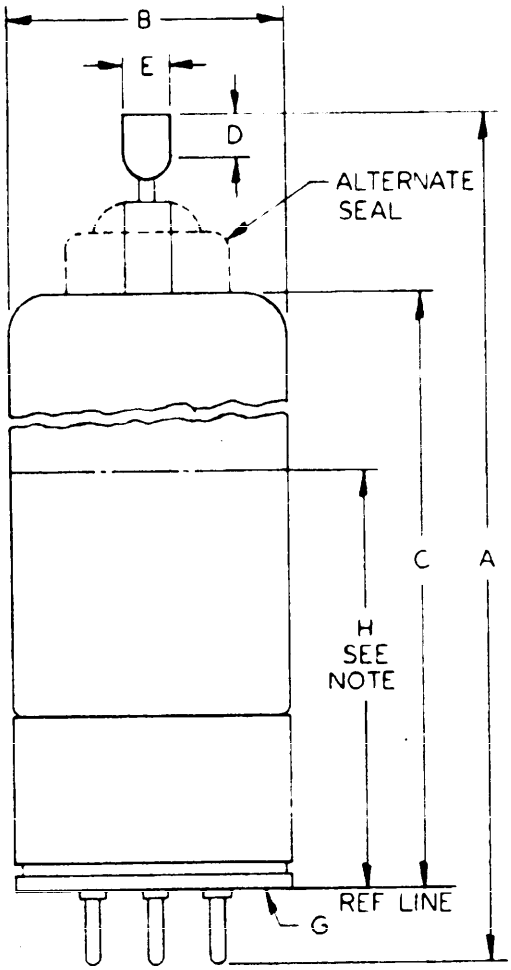
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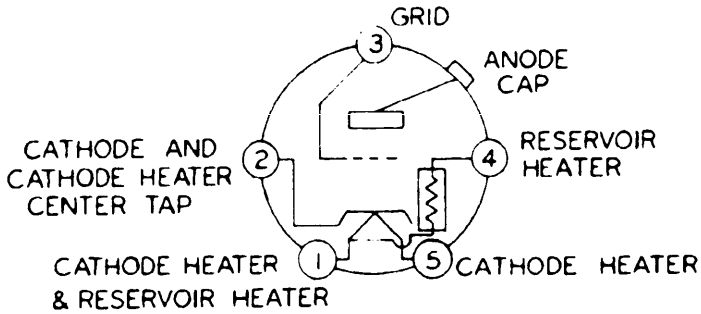
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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	11.5 (292.1)	12.5 (317.5)
B	- - - - - -	3.313 (84.15)
C	8.00 (203.2)	9.00 (228.6)
H	- - - - - -	5.00 (127.0)
Quality conformance inspection, part 3 (SEE NOTE 7)		
E	.559 (14.20)	.573 (14.55)
G	Base: A5-98 (EIA)	
Reference dimension		
D	.500 (12.70)	



BASE CONNECTIONS

NOTE:
Dimension H locates the center line of the clamping area. Clamping shall be confined to 1.5 inches above and below this line.

FIGURE 1. Outline drawing of electron tube type 5949A.

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